
Auto Driving RL

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1 Background and Motivation

We study reinforcement learning for closed-loop autonomous driving in an interactive multi-agent environment. Let $(\mathcal{X}, \mathcal{U}, P, r, \gamma)$ be an episodic MDP with observation $x_t \in \mathcal{X}$ and continuous control $u_t \in \mathcal{U} \subset \mathbb{R}^2$. The objective is

$$\max_{\theta} \mathbb{E}_{\tau \sim \pi_{\theta}} \left[\sum_{t=0}^{T-1} \gamma^t r(x_t, u_t, x_{t+1}) \right], \quad \tau = (x_0, u_0, \dots, x_T). \quad (1)$$

Driving rewards are typically safety-dominated:

$$r_t = r^{\text{prog}}(x_t, x_{t+1}) - \kappa c(x_t, u_t), \quad (2)$$

where c aggregates safety costs (collision/off-road/rule violations) and $\kappa > 0$. This induces (i) unsafe naive exploration and (ii) weak identifiability of safety margins under similar returns.

2 TREX as a Sampling Policy Primitive

Let $\pi_{\theta}(u \mid x)$ be the ego policy. TREX can be viewed as learning under a proposal sampler that (i) enforces feasibility and (ii) tilts sampling by a dense potential.

Feasibility support. Define $\mathcal{F}(x_t, u_t) \in \{0, 1\}$. TREX restricts trajectory support to feasible prefixes:

$$\text{supp}(\tilde{\pi}) \subseteq \{(x_t, u_t) : \mathcal{F}(x_t, u_t) = 1, \forall t\}. \quad (3)$$

Potential-tilted proposal. TREX samples controls from

$$q_{\theta}(u_t \mid x_t) \propto \pi_{\theta}(u_t \mid x_t) \exp(\lambda \Phi(x_t, u_t)), \quad (4)$$

where Φ is a dense guidance potential.

However, in driving, feasibility is predictive and interaction-dependent:

$$\mathcal{F}(x_t, u_t) = \mathbf{1}[g(x_t, u_t, \omega_t) \leq 0], \quad \omega_t \sim P_{\omega}(\cdot \mid x_t), \quad (5)$$

so any estimate $\hat{\mathcal{F}}$ induces distribution bias $q_{\theta}(\tau \mid \hat{\mathcal{F}}) \neq q_{\theta}(\tau \mid \mathcal{F})$. Moreover, risk-dominant tilting can admit conservative fixed points: if $\Phi(x, u) \approx -\text{risk}(x, u)$ then $\arg \max_u \Phi$ may satisfy $\Delta_{\text{prog}} \approx 0$ (stalling), yielding mode collapse. Finally, shaped returns can be weakly identifiable w.r.t. safety margins:

$$\exists \tau_1, \tau_2 : R_{\text{env}}(\tau_1) \approx R_{\text{env}}(\tau_2) \text{ but } \text{risk}(\tau_1) \ll \text{risk}(\tau_2), \quad (6)$$

so sampling must be coupled with a label-free regularizer that separates *safe-progress* behavior modes.

22 **3 Proposed: Topo-MCPO — Topology-regularized Meaning-Consensus** 23 **Policy Optimization for Auto-Driving**

24 **3.1 Key Shift in Notation and Novelty**

25 We propose a new RL framework, **Topo-MCPO**. Topo-MCPO introduces a closed-loop coupling
26 between:

- 27 1. Topology-tilted, constraint-supported sampling, and
 - 28 2. Quality-weighted behavioral meaning consensus,
- 29 through a shared behavioral manifold and an entropy thermostat that adaptively governs exploration
30 vs. mode-collapse in safety-critical driving.

31 **3.2 Driving Objective**

32 We use the MDP objective in (1) with reward decomposition (2).

33 **3.3 (A) Constraint-Supported, Topology-Tilted Group Sampling**

34 Topo-MCPO samples a group of rollouts per scenario, but the sampling distribution is cluster-aware
35 and collapse-resistant.

36 **Feasibility predicate for driving.** We define

$$\mathcal{F}(x_t, u_t) = \mathbf{1}[\text{drivable}(x_{t+1})] \cdot \mathbf{1}[\text{dyn}(x_t, u_t)] \cdot \mathbf{1}[\text{TTC}(x_t, u_t) > \epsilon], \quad (7)$$

37 where dyn checks kinematic/actuation limits and TTC is a short-horizon time-to-collision surrogate.

38 **Risk-progress topology potential.** We decompose the potential into route topology and risk
39 topology:

$$\Phi(x_t, u_t) = \Phi_{\text{route}}(x_t, u_t) - \beta \Phi_{\text{risk}}(x_t, u_t), \quad (8)$$

40 where Φ_{route} can be a graph distance-to-goal decrease (route/lane graph potential), and Φ_{risk} can be
41 predicted near-collision/off-road risk under short-horizon rollout.

42 **Collapse-resistant manifold novelty tilt.** Let $\tau^{(g)}$ be the g -th sampled trajectory in the group, and
43 define a behavior embedding

$$z(\tau) = \Psi(\text{progress profile, interaction modes, risk signature, comfort}) \in \mathbb{R}^d. \quad (9)$$

44 We maintain an exponential-moving-average (EMA) density $\hat{p}(z)$ over past sampled embeddings.
45 Topo-MCPO samples each rollout using an additional novelty tilt:

$$q_{\theta}^{\text{Topo}}(\tau) \propto \left[\prod_t \pi_{\theta}(u_t | x_t) \exp(\lambda \Phi(x_t, u_t)) \right] \cdot \exp(\eta \underbrace{(-\log \hat{p}(z(\tau)))}_{\text{novelty}}), \quad (10)$$

46 subject to $\mathcal{F}(x_t, u_t) = 1$ for all steps. This term prevents a safe but degenerate attractor (e.g., “stop
47 forever”) by ensuring the sampler continues to cover multiple feasible behavior modes.

48 **3.4 (B) Meaning-Consensus Reward with Quality Weighting**

49 Given a scenario s (traffic scene) we sample a group $\{\tau_1, \dots, \tau_G\}$ using (10). We cluster the behavior
50 embeddings $\{z(\tau_g)\}$ into M clusters $\{C_1, \dots, C_M\}$, and estimate cluster mass:

$$\hat{\rho}(C_m | s) = \frac{|C_m|}{G}. \quad (11)$$

51 **Novel: quality-weighted consensus mass.** Vanilla consensus (“majority wins”) is unsafe in driving
 52 because it can reward consistent stagnation. Topo-MCPO assigns each cluster a *quality score*:

$$Q(C_m) = \sigma(a_1 \overline{\Delta \text{prog}}(C_m) - a_2 \overline{\text{risk}}(C_m) - a_3 \overline{\text{viol}}(C_m)), \quad (12)$$

53 where $\overline{\Delta \text{prog}}$ is mean route progress, $\overline{\text{risk}}$ is mean risk proxy (e.g., min TTC or predicted collision
 54 probability), and $\overline{\text{viol}}$ aggregates rule violations; $\sigma(\cdot)$ is a squashing function.

55 For each trajectory $\tau_g \in C(\tau_g)$, we define the meaning-consensus intrinsic reward:

$$r^{\text{mc}}(\tau_g) = \hat{\rho}(C(\tau_g) | s) \cdot Q(C(\tau_g)). \quad (13)$$

56 This is the second coupling: consensus is quality-gated by safety and progress, so the policy is pulled
 57 toward dominant safe-progress modes, not merely consistent modes.

58 3.5 (C) Entropy Thermostat: Adaptive Control of Diversity vs. Commitment

59 Define the group meaning entropy:

$$\mathcal{H}(s) = - \sum_{m=1}^M \hat{\rho}(C_m | s) \log \hat{\rho}(C_m | s). \quad (14)$$

60 Instead of fixed thresholding, Topo-MCPO uses an entropy thermostat that adjusts (η, α) online:

$$\eta(s) = \eta_0 + k_\eta(\mathcal{H}^* - \mathcal{H}(s)), \quad \alpha(s) = \alpha_0 + k_\alpha(\mathcal{H}(s) - \mathcal{H}^*), \quad (15)$$

61 where $\mathcal{H}^*$ is a target diversity level. If entropy is too low (mode collapse), novelty tilt increases; if
 62 too high (indecision/noise), intrinsic consensus weight increases. The exploration/exploitation is
 63 controlled per scenario, which is crucial under heterogeneous traffic difficulty.

64 3.6 (D) Final Objective: Group-Relative RL with Topo-MCPO Returns

65 Let the environment return be $R_{\text{env}}(\tau) = \sum_t \gamma^t r_t$. Topo-MCPO defines the augmented trajectory
 66 score:

$$\tilde{R}(\tau_g) = R_{\text{env}}(\tau_g) + \alpha(s) r^{\text{mc}}(\tau_g). \quad (16)$$

67 We compute group-relative advantages:

$$A_g = \frac{\tilde{R}(\tau_g) - \frac{1}{G} \sum_{i=1}^G \tilde{R}(\tau_i)}{\text{Std}(\{\tilde{R}(\tau_i)\}_{i=1}^G) + \varepsilon}. \quad (17)$$

68 Then we optimize a clipped surrogate (PPO/GRPO-compatible):

$$\max_{\theta} \mathbb{E} \left[\frac{1}{G} \sum_{g=1}^G \min \left(w_{\theta}(\tau_g) A_g, \text{clip}(w_{\theta}(\tau_g), 1 - \epsilon, 1 + \epsilon) A_g \right) - \mu \text{KL}(\pi_{\theta} \| \pi_{\text{ref}}) \right], \quad (18)$$

69 where $w_{\theta}(\tau_g) = \prod_t \frac{\pi_{\theta}(u_t | x_t)}{\pi_{\theta_{\text{old}}}(u_t | x_t)}$ and μ can be annealed.